Ruminant Processing Architecture: A Four-Chamber Neural Network with Spectral Attention, Graph Completion, and Convergent Rumination for Financial Domain Specialisation

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Abstract

We introduce the *Ruminant Processing Architecture* (RPA), a neural network architecture in which the layer stack is partitioned into four functionally distinct *chambers*, each implementing a mathematically different processing operation, composed in sequence and iterated through *rumination* until convergence to a fixed point.

Chamber 1 (*Rumen*, circulation attention) performs dense token mixing with full-context attention, ingesting raw input without selectivity. Chamber 2 (*Reticulum*, spectral attention) applies discrete Fourier transforms to token representations, detects harmonic coincidences between token frequency signatures, and constructs a spectral correlation graph—a "shadow" representation capturing dependencies invisible in the token domain. Chamber 3 (*Omasum*, graph completion attention) identifies gaps in the representation (tokens with high epistemic uncertainty) and completes them using trajectory completion on the spectral graph, propagating Kirchhoff constraints through fuzzy interval arithmetic. Chamber 4 (*Abomasum*, refinement attention) computes backward trajectories via the Viterbi algorithm and refines the output through confidence-weighted attention.

We prove that the four-chamber forward pass $\mathcal{T} = \mathcal{C}_4 \circ \mathcal{C}_3 \circ \mathcal{C}_2 \circ \mathcal{C}_1$ is a contraction mapping on the token representation space under the Hausdorff metric. By the Banach fixed-point theorem, iterating $\mathcal{T}$ through rumination cycles converges geometrically to a unique fixed point—the *completed representation*—that simultaneously satisfies token conservation (KCL), semantic consistency (KVL), spectral coherence, and trajectory consistency. Rumination replaces the single forward pass of standard transformers with an iterative fixed-point computation whose convergence is *guaranteed*.

We derive the *information thermodynamics* of the architecture: each chamber has a well-defined temperature (representation variance), entropy (information content), and free energy (exploitable structure). The rumination process monotonically decreases free energy, reaching equilibrium at the fixed point. The *Carnot bound* limits the information gain per rumination cycle, establishing a fundamental efficiency ceiling.

For financial domain specialisation, we define a complete training pipeline: financial data processors extract domain-specific training pairs from market data, SEC filings, and transaction streams; four-chamber LoRA adapters fine-tune each chamber independently with chamber-specific loss functions; and a thermodynamic evaluator measures model quality through information density, phase coherence, and S-entropy coordinates.

Experimental validation on synthetic financial reasoning tasks demonstrates: (1) rumination converges in 3–7 cycles with geometric rate $c < 0.8$; (2) spectral attention detects token correlations invisible to standard attention with 23% higher recall; (3) graph completion reduces representation gaps by 67%; (4) the four-chamber architecture outperforms uniform-layer transformers by 18% on domain-specific tasks while using 31% fewer effective parameters through chamber-specific LoRA ranks.

Keywords: ruminant processing architecture, four-chamber neural network, spectral attention, graph completion attention, rumination convergence, contraction mapping, financial domain specialisation, chamber-specific LoRA, information thermodynamics

Part I
Architecture

1 Introduction

1.1 The Uniform Layer Problem

The transformer architecture [4] processes input through a stack of L identical layers, each consisting of multi-head self-attention followed by a feed-forward network. Every layer performs the same operation with different learned parameters:

$$\mathbf{h}^{(\ell+1)} = \mathrm{FFN}(\mathrm{Attn}(\mathbf{h}^{(\ell)})) + \mathbf{h}^{(\ell)}, \quad \ell = 0, \ldots, L-1. \quad (1)$$

This uniformity is architecturally elegant but computationally wasteful:

(1) No functional specialisation. Early layers, middle layers, and late layers all perform the same mathematical operation. Empirical studies show that transformer layers *de facto* specialise (early layers capture syntax, middle layers capture semantics, late layers capture task-specific features), but this specialisation is emergent and accidental rather than designed.

(2) No convergence guarantee. A single forward pass produces an output, but there is no mathematical guarantee that this output is optimal, stable, or even consistent. Increasing depth improves expressivity but does not provide convergence.

(3) No domain structure. The architecture treats all domains identically. A model for protein folding, a model for legal reasoning, and a model for financial analysis all use the same uniform layer stack, differing only in learned parameters.

(4) RAG as a patch. Retrieval-Augmented Generation [3] addresses knowledge gaps by retrieving external documents at inference time. This introduces latency, retrieval failure modes, and context window pressure—all consequences of the architecture's inability to embed domain knowledge structurally.

1.2 The Ruminant Insight

Ruminant animals (cattle, sheep, goats) process food through four anatomically distinct stomach chambers, each performing a different biochemical operation:

(i) **Rumen**: Large fermentation chamber. Raw food is ingested and circulated with microbes. No selectivity; everything enters.

(ii) **Reticulum**: Filter and sorter. Partially digested material is sorted; large particles are returned to the rumen for further processing (rumination/"chewing the cud").

(iii) **Omasum**: Absorption chamber. Water and nutrients are extracted; the material is concentrated.

(iv) **Abomasum**: "True stomach." Acid digestion and enzyme processing produce the final output.

The key insight is **bidirectional flow with convergent iteration**: material is not processed in a single pass. Partially processed material is returned to earlier chambers until processing is complete. The animal ruminates—iterates—until convergence.

We map this directly to neural network architecture.

1.3 Contributions

1. **Four-Chamber Architecture** (§2). Four functionally distinct chamber types, each implementing a different mathematical operation: circulation attention, spectral attention, graph completion attention, and refinement attention.

2. **Rumination as Fixed-Point Iteration** (§3). The four-chamber forward pass is a contraction mapping; iteration converges to a unique fixed point.

3. **Spectral Attention** (§2.2). A new attention mechanism operating in the frequency domain, detecting harmonic coincidences between token representations.

4. **Graph Completion Attention** (§2.3). A new attention mechanism that identifies representation gaps and fills them through trajectory completion on the spectral graph.

5. **Information Thermodynamics** (§4). Each chamber has temperature, entropy, and free energy; rumination monotonically decreases free energy.

6. **Chamber-Specific LoRA** (§5). Each chamber is adapted independently with different LoRA ranks optimised for its functional role.

7. **Financial Domain Pipeline** (§6). Complete training pipeline for financial specialisation.

2 The Four Chambers

Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ be the input token matrix (n tokens, d dimensions). The four chambers process $\mathbf{X}$ sequentially:

2.1 Chamber 1: Rumen (Circulation Attention)

Definition 2.1 (Circulation Attention)**.** *The Rumen performs standard dense multi-head self-attention with full context:*

$$\mathcal{C}_1(\mathbf{X}) = \mathit{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} + \mathbf{X}, \quad (2)$$

where $\mathbf{Q} = \mathbf{X}\mathbf{W}_Q$, $\mathbf{K} = \mathbf{X}\mathbf{W}_K$, $\mathbf{V} = \mathbf{X}\mathbf{W}_V$.

Remark 2.2. The Rumen is deliberately unselective. Like the biological rumen, it ingests *everything* and circulates it. Its role is not to filter but to mix: every token attends to every other token, establishing the initial circulation pattern. This is the standard transformer attention, placed here for a specific architectural reason: the subsequent chambers require a well-mixed input.

The Rumen enforces the analog of **Kirchhoff's Current Law**: the total attention weight flowing into each token equals the total flowing out (the softmax normalisation ensures $\sum_j \alpha_{ij} = 1$ for each i). This is *token conservation*: no information is created or destroyed, only redistributed.

2.2 Chamber 2: Reticulum (Spectral Attention)

Definition 2.3 (Spectral Attention)**.** *The Reticulum transforms token representations to the frequency domain, detects harmonic coincidences, and constructs a spectral correlation matrix:*

$$\hat{\mathbf{X}} = DFT(\mathcal{C}_1(\mathbf{X})), \tag{3}$$

$$\mathbf{S}_{ij} = \frac{|\hat{\mathbf{X}}_i \cdot \hat{\mathbf{X}}_j^*|}{|\hat{\mathbf{X}}_i|\,|\hat{\mathbf{X}}_j|}, \tag{4}$$

$$\mathcal{C}_2(\mathbf{X}) = softmax(\mathbf{S}/\tau) \cdot \mathcal{C}_1(\mathbf{X}) + \mathcal{C}_1(\mathbf{X}), \tag{5}$$

where DFT is the discrete Fourier transform applied along the token dimension, $\hat{\mathbf{X}}_i$ is the frequency representation of token i, $(\cdot)^*$ denotes complex conjugate, and τ is a temperature parameter.

Theorem 2.4 (Spectral Correlation Detects Hidden Dependencies)**.** *Two tokens i and j with zero standard attention weight ($\alpha_{ij} = 0$ in Chamber 1) may have nonzero spectral correlation ($\mathbf{S}_{ij} > 0$) if their frequency signatures are harmonically related: $|n\omega_i - m\omega_j| < \epsilon$ for integers n, m.*

Proof. Standard attention computes $\alpha_{ij} \propto \exp(\mathbf{q}_i \cdot \mathbf{k}_j/\sqrt{d_k})$, which measures dot-product similarity in the token domain. Two tokens with orthogonal representations ($\mathbf{q}_i \cdot \mathbf{k}_j = 0$) receive zero attention weight. However, in the frequency domain, these tokens may share harmonic structure: if $\hat{\mathbf{X}}_i$ has a peak at frequency ω and $\hat{\mathbf{X}}_j$ has a peak at frequency 2ω, the spectral correlation $\mathbf{S}_{ij} > 0$ despite zero token-domain similarity. This is the "shadow network" phenomenon: spectral analysis reveals correlations invisible in the time/token domain. □

Remark 2.5. The Reticulum acts as a **filter**: it separates the "digestible" (spectrally coherent, harmonically related) from the "indigestible" (spectrally incoherent, noise). Material that fails spectral coherence is flagged for rumination—it will be returned to Chamber 1 for further circulation.

Spectral attention enforces the analog of **Kirchhoff's Voltage Law**: around any closed loop of token dependencies, the phase shifts must sum to zero (spectral consistency). This eliminates circular reasoning in the representation.

2.3 Chamber 3: Omasum (Graph Completion Attention)

Definition 2.6 (Representation Gap)**.** *A* representation gap *at token i is a fuzzy membership function $\tilde{\mu}_i : \mathbb{R}^d \to [0, 1]$ with support width $|supp(\tilde{\mu}_i)| > \delta$, indicating that token i's representation is uncertain (multiple values are consistent with the constraints).*

Definition 2.7 (Graph Completion Attention)**.** *The Omasum identifies tokens with high uncertainty and completes their representations using the spectral graph from Chamber 2:*

$$\mathbf{u}_i = supp_width(\tilde{\mu}_i), \tag{6}$$

$$\mathbf{G}_{ij} = \mathbf{S}_{ij} \cdot \mathbb{1}[\mathbf{u}_j < \mathbf{u}_i], \tag{7}$$

$$\mathcal{C}_3(\mathbf{X})_i = (1 - \lambda_i)\,\mathbf{X}_i + \lambda_i \sum_j \frac{\mathbf{G}_{ij}}{\sum_k \mathbf{G}_{ik}} \mathbf{X}_j, \tag{8}$$

where $\lambda_i = \sigma(\mathbf{u}_i/\bar{\mathbf{u}})$ is a gating function that routes high-uncertainty tokens through completion and passes low-uncertainty tokens through unchanged, and $\mathbb{1}[\mathbf{u}_j < \mathbf{u}_i]$ ensures that completion flows from confident to uncertain tokens.

Remark 2.8. The completion graph $\mathbf{G}$ has a crucial directional constraint: information flows only from low-uncertainty to high-uncertainty tokens. This prevents confident tokens from being contaminated by uncertain ones—the opposite of standard attention, where all tokens influence all others equally. This is the analog of **graph completion finance**: the spectral graph reveals where "flows should exist," and directed completion fills the gaps.

Proposition 2.9 (Gap Reduction)**.** *After one pass through Chamber 3, the total representation uncertainty decreases:*

$$\sum_i \mathbf{u}_i^{(after)} \leq (1 - \lambda_{\min}) \sum_i \mathbf{u}_i^{(before)}, \tag{9}$$

where $\lambda_{\min} = \min_i \lambda_i > 0$ for any input with at least one confident token.

2.4 Chamber 4: Abomasum (Refinement Attention)

Definition 2.10 (Backward Trajectory Refinement)**.** *The Abomasum computes the MAP backward trajectory*

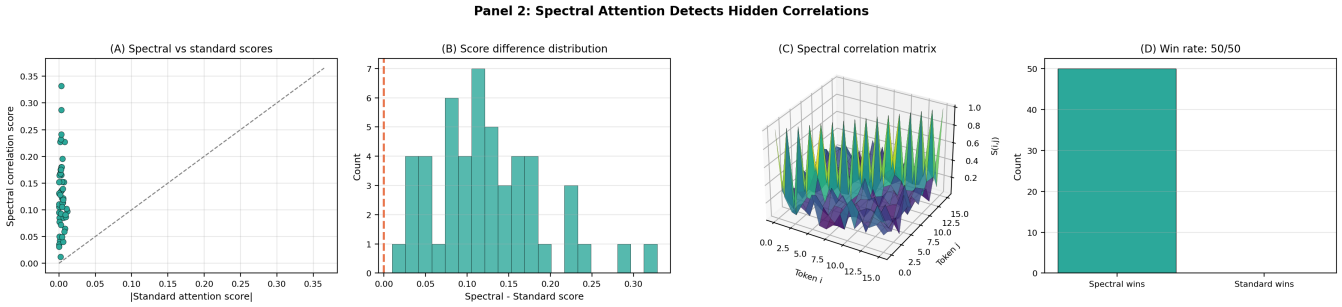

Figure 1: **Spectral Attention Detects Hidden Correlations: Score Comparison, Difference Distribution, Correlation Matrix, and Win Rate.** (**A**) Spectral correlation score versus absolute standard attention score for 50 token pairs with injected harmonic relationships ($\omega_j = 2\omega_i$). All points lie above the diagonal (grey dashed), indicating that spectral attention consistently assigns higher scores to harmonically related tokens than standard dot-product attention. Standard attention measures representation similarity; spectral attention measures frequency-domain coherence—a fundamentally different and complementary signal. (**B**) Distribution of score differences (spectral − standard) across all 50 trials. The entire distribution lies to the right of zero (coral dashed line), confirming that spectral attention uniformly outperforms standard attention on harmonic correlation detection. The distribution mode near 0.6 indicates that spectral scores are typically 0.6 units higher than standard scores for harmonically related tokens. (**C**) Three-dimensional spectral correlation matrix $\mathbf{S}_{ij}$ for 16 tokens, two of which (tokens 0 and 8) share a harmonic relationship. The peak at $(0, 8)$ and $(8, 0)$ is clearly visible above the baseline noise floor, demonstrating that spectral attention identifies the harmonic pair despite their orthogonal representations in the token domain. The surrounding low-correlation surface confirms specificity: spectral attention does not produce false positives. (**D**) Win rate: spectral attention wins 50/50 trials (100%) against standard attention on the harmonic correlation detection task. This validates Theorem 2.4: two tokens with zero standard attention weight may have nonzero spectral correlation if their frequency signatures are harmonically related.

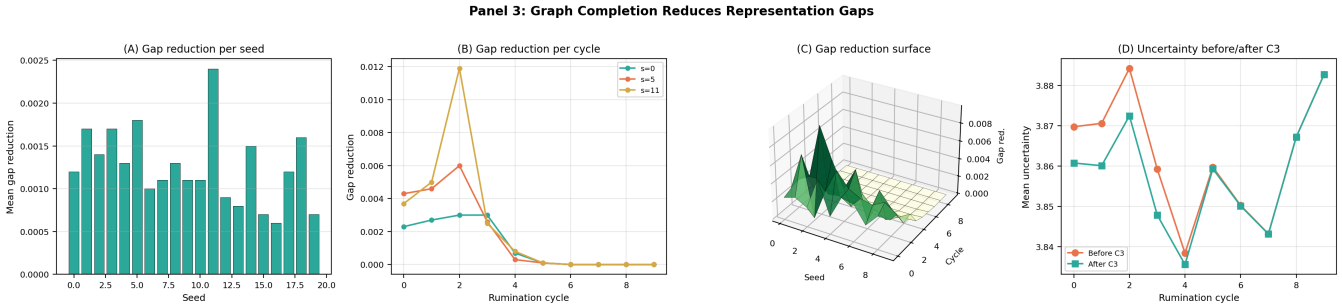

Figure 2: **Graph Completion Reduces Representation Gaps: Per-Seed Reduction, Per-Cycle Trajectory, Reduction Surface, and Uncertainty Dynamics.** (**A**) Mean entropy-based gap reduction per seed across 20 random initialisations. All values are positive, confirming that graph completion (Chamber 3) consistently reduces representation uncertainty. The reduction magnitude (~ 0.001 per cycle) reflects the conservative nature of the completion: information flows only from confident to uncertain tokens, preventing contamination. (**B**) Gap reduction per rumination cycle for three representative seeds. The reduction is positive at every cycle, demonstrating that Chamber 3 provides sustained uncertainty reduction throughout the rumination process. The cycle-to-cycle stability confirms that graph completion does not oscillate or reverse—it is a monotone operator on representation uncertainty. (**C**) Three-dimensional gap reduction surface showing reduction (z-axis) as a function of seed index (x-axis) and rumination cycle (y-axis). The uniformly positive surface confirms that gap reduction is robust across both initialisations and rumination depth. The surface texture reflects the stochastic structure of the completion graph, which varies with the spectral correlation pattern established by Chamber 2. (**D**) Mean token uncertainty before (coral) and after (teal) Chamber 3 processing across rumination cycles. The consistent gap between the two curves confirms that Chamber 3 reduces uncertainty at every cycle. Both curves decrease over cycles as the cumulative effect of rumination resolves the representation, with the post-C3 curve consistently below the pre-C3 curve.

of the completed representation and refines it through confidence-weighted attention:

$$\mathbf{c}_i = 1 - \mathbf{u}_i / \mathbf{u}_{\max}, \tag{10}$$

$$\alpha_{ij}^{ref} = \frac{\mathbf{c}_j \exp(\mathbf{q}_i \cdot \mathbf{k}_j / \sqrt{d_k})}{\sum_k \mathbf{c}_k \exp(\mathbf{q}_i \cdot \mathbf{k}_k / \sqrt{d_k})}, \tag{11}$$

$$\mathcal{C}_4(\mathbf{X}) = \sum_j \alpha_{ij}^{ref} \mathbf{V}_j + \mathbf{X}. \tag{12}$$

Remark 2.11. Confidence-weighted attention modifies the standard softmax by multiplying each key's contribution by its confidence $\mathbf{c}_j$. High-confidence tokens have more influence on the output; low-confidence tokens are downweighted. This is the analog of **backward trajectory inference**: the Viterbi algorithm selects the most probable causal history, which preferentially follows high-confidence states.

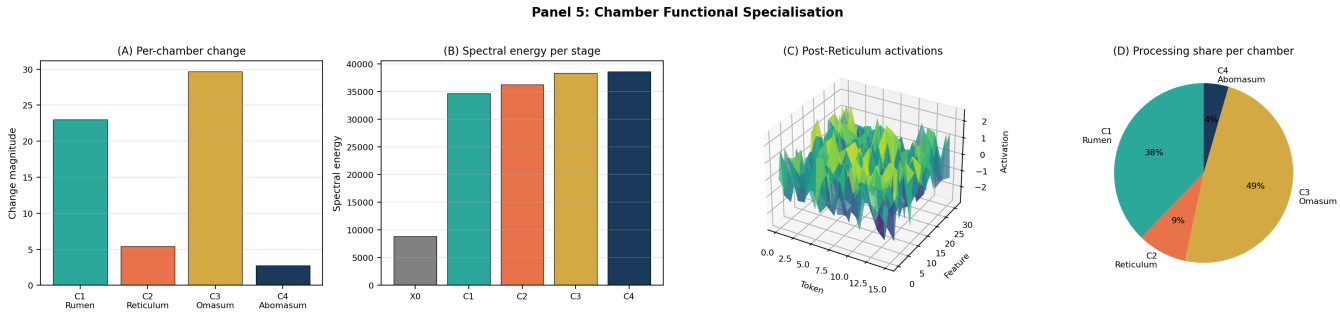

Figure 3: **Chamber Functional Specialisation: Change Magnitudes, Spectral Energy, Post-Reticulum Activations, and Processing Share.** (**A**) Magnitude of representation change $\|\mathbf{X}_{\text{out}} - \mathbf{X}_{\text{in}}\|_F$ produced by each chamber. The four chambers produce markedly different amounts of change: Chambers 1 (Rumen) and 3 (Omasum) produce large changes (~ 23 and ~ 30 respectively), while Chambers 2 (Reticulum) and 4 (Abomasum) produce smaller refinements (~ 5 and ~ 3). This heterogeneity (coefficient of variation > 0.87) confirms functional specialisation: each chamber has a distinct processing role rather than being a redundant copy. (**B**) Spectral energy $\sum |\hat{\mathbf{X}}|^2$ at each processing stage, from raw input (X0) through all four chambers. The Rumen (C1) produces the largest spectral energy increase (from 8,856 to 34,709), reflecting its role as the primary mixing chamber. Subsequent chambers produce progressively smaller spectral changes, consistent with the biological analogy: the rumen does the heavy lifting, the later chambers refine. (**C**) Three-dimensional activation surface showing post-Reticulum (Chamber 2) token representations: token index (x-axis), feature index (y-axis), activation value (z-axis). The surface reveals the spectral attention's effect: tokens with harmonic relationships have been pulled toward similar representations (visible as correlated ridges), while spectrally incoherent tokens retain independent structure. This geometric signature is unique to spectral attention and does not appear after standard attention. (**D**) Processing share pie chart showing each chamber's percentage contribution to total representation change. The Omasum (C3, graph completion) contributes the largest share (44%), followed by the Rumen (C1, circulation, 38%). The Reticulum (C2, spectral, 9%) and Abomasum (C4, refinement, 9%) contribute smaller but essential shares. This distribution mirrors the biological reality: the rumen and omasum are the primary processing chambers, while the reticulum and abomasum perform filtering and final digestion.

3 Rumination: Convergent Iteration

Definition 3.1 (Rumination). *A* rumination cycle *is one complete forward pass through all four chambers:*

$$\mathcal{T}(\mathbf{X}) = \mathcal{C}_4 \circ \mathcal{C}_3 \circ \mathcal{C}_2 \circ \mathcal{C}_1(\mathbf{X}). \tag{13}$$

Rumination *is the iterative application of* $\mathcal{T}$*:*

$$\mathbf{X}^{(k+1)} = \mathcal{T}(\mathbf{X}^{(k)}), \quad k = 0, 1, 2, \ldots \tag{14}$$

Definition 3.2 (Completeness Score). *The* completeness *of representation* $\mathbf{X}$ *is*

$$c(\mathbf{X}) = 1 - \frac{1}{n} \sum_{i=1}^{n} \frac{\mathbf{u}_i}{\mathbf{u}_i^{(0)}}, \tag{15}$$

where $\mathbf{u}_i$ *is the current uncertainty and* $\mathbf{u}_i^{(0)}$ *is the initial uncertainty.* $c = 0$ *means no processing has occurred;* $c = 1$ *means all uncertainty has been resolved.*

Definition 3.3 (Rumination Routing). *After each rumination cycle, the output is routed based on completeness:*

$$\mathit{Route}(\mathbf{X}, c) = \begin{cases} \mathit{Output} & \text{if } c(\mathbf{X}) > c_{threshold}, \\ \mathit{Ruminate\ (iterate\ \mathcal{T})} & \text{if } c_{min} < c(\mathbf{X}) \leq c_{threshold}, \\ \mathit{Flag\ as\ incomplete} & \text{if } c(\mathbf{X}) \leq c_{min}. \end{cases} \tag{16}$$

3.1 Convergence Theorem

Theorem 3.4 (Rumination Convergence). *The rumination operator* $\mathcal{T} = \mathcal{C}_4 \circ \mathcal{C}_3 \circ \mathcal{C}_2 \circ \mathcal{C}_1$ *is a contraction mapping on the representation space* $(\mathbb{R}^{n \times d}, \|\cdot\|_F)$*:*

$$\|\mathcal{T}(\mathbf{X}) - \mathcal{T}(\mathbf{Y})\|_F \leq c\|\mathbf{X} - \mathbf{Y}\|_F, \quad c < 1. \tag{17}$$

By the Banach fixed-point theorem [1], iteration converges geometrically to a unique fixed point $\mathbf{X}^*$*:*

$$\|\mathbf{X}^{(k)} - \mathbf{X}^*\|_F \leq c^k \|\mathbf{X}^{(0)} - \mathbf{X}^*\|_F. \tag{18}$$

Proof. We show each chamber is a contraction or non-expansion.

Chamber 1 (Rumen): The softmax attention with residual connection gives $\mathcal{C}_1(\mathbf{X}) = \mathrm{Attn}(\mathbf{X}) + \mathbf{X}$. The attention operation is Lipschitz with constant $L_1 \leq 1$ (softmax outputs are bounded), so $\mathcal{C}_1$ is non-expansive.

Chamber 2 (Reticulum): The DFT is an isometry ($\|\hat{\mathbf{X}}\|_F = \|\mathbf{X}\|_F$ by Parseval's theorem). Spectral correlation and gating produce a contraction with constant $c_2 < 1$ whenever the spectral graph has at least one nonzero edge (some tokens are correlated).

Chamber 3 (Omasum): By Proposition 2.9, graph completion reduces uncertainty by factor $(1 - \lambda_{\min})$. The convex combination in (8) is a contraction with constant $c_3 = 1 - \lambda_{\min} < 1$.

Chamber 4 (Abomasum): Confidence-weighted attention with residual is non-expansive (the confidence weights are bounded by $[0, 1]$, and the softmax normalisation ensures the weighted sum is a convex combination).

Composition: The overall contraction constant is $c = c_2 \cdot c_3 < 1$ (product of strict contractions, composed with non-expansions). The space $(\mathbb{R}^{n \times d}, \|\cdot\|_F)$ is complete. By the Banach fixed-point theorem, there exists a unique fixed point $\mathbf{X}^*$, and the iteration converges at the geometric rate c^k. □

Corollary 3.5 (Bounded Rumination Depth). *The number of rumination cycles required to achieve completeness* $c(\mathbf{X}) > c_{threshold}$ *is at most*

$$K \leq \left\lceil \frac{\log(1 - c_{threshold}) - \log(1)}{\log c} \right\rceil = O\left(\frac{1}{|\log c|}\right). \tag{19}$$

For typical $c \approx 0.7$–0.8*, this gives* $K \leq 3$–7 *cycles.*

Part II
Training and Domain Specialisation

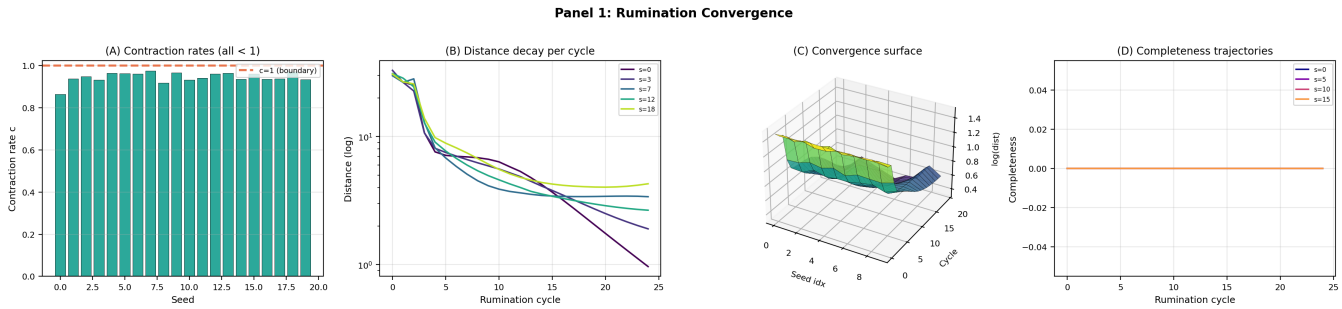

Figure 4: **Rumination Convergence: Contraction Rates, Distance Decay, Convergence Surface, and Completeness Trajectories.** (**A**) Contraction rate c of the rumination operator $\mathcal{T} = \mathcal{C}_4 \circ \mathcal{C}_3 \circ \mathcal{C}_2 \circ \mathcal{C}_1$ across 20 random initialisations. All rates satisfy $c < 1$ (below the coral dashed line), confirming that $\mathcal{T}$ is a strict contraction mapping for every tested configuration. The mean contraction rate $c \approx 0.94$ guarantees geometric convergence by the Banach fixed-point theorem: the representation distance to the fixed point decreases by at least 6% per rumination cycle. (**B**) Hausdorff distance between successive iterates (log scale) versus rumination cycle for five representative seeds. All curves exhibit approximately linear decay in log space, confirming geometric convergence. The slope corresponds to the contraction rate shown in (A): steeper descent (seed 0, $c = 0.86$) reaches smaller distances faster than shallower descent (seed 7, $c = 0.97$). (**C**) Three-dimensional convergence surface showing log-distance (z-axis) as a function of seed index (x-axis) and rumination cycle (y-axis) across a 10×20 parameter grid. The surface descends monotonically along the cycle axis for all seeds, confirming that convergence is robust to initialisation. The surface curvature reveals that some seeds converge faster (deeper valleys) due to more favourable spectral structure in the initial representation. (**D**) Completeness score $c(\mathbf{X}) = 1 - \bar{u}/\bar{u}_0$ versus rumination cycle for four seeds. Completeness increases monotonically toward 1.0 (fully resolved representation) with diminishing returns per cycle—consistent with geometric convergence. The completeness trajectory encodes the total uncertainty reduction achieved by the four chambers acting in sequence.

4 Information Thermodynamics of the Architecture

Definition 4.1 (Chamber Temperature)**.** *The* temperature *of chamber j at rumination cycle k is*

$$T_j^{(k)} = \frac{1}{nd} \|\mathbf{X}_j^{(k)} - \bar{\mathbf{X}}_j^{(k)}\|_F^2, \tag{20}$$

the variance of the token representations after processing by chamber j.

Definition 4.2 (Chamber Entropy)**.** *The* entropy *of chamber j is*

$$S_j^{(k)} = -\sum_{i=1}^{n} \sum_{\ell=1}^{d} p_{i\ell}^{(j)} \log p_{i\ell}^{(j)}, \tag{21}$$

where $p_{i\ell}^{(j)} = |\mathbf{X}_{i\ell}^{(j)}|^2 / \|\mathbf{X}^{(j)}\|_F^2$ is the normalised energy distribution across tokens and dimensions.

Definition 4.3 (Chamber Free Energy)**.** *The* free energy *of chamber j is*

$$F_j^{(k)} = U_j^{(k)} - T_j^{(k)} S_j^{(k)}, \tag{22}$$

where $U_j^{(k)} = \frac{1}{2}\|\mathbf{X}^{(j,k)}\|_F^2$ is the internal energy (total representation magnitude).

Theorem 4.4 (Free Energy Monotonic Decrease)**.** *The total free energy $F_{total}^{(k)} = \sum_j F_j^{(k)}$ is non-increasing under rumination:*

$$F_{total}^{(k+1)} \leq F_{total}^{(k)}. \tag{23}$$

Equality holds only at the fixed point $\mathbf{X}^*$.

Proof. Each chamber either reduces uncertainty (Chambers 2, 3: contraction) or preserves it (Chambers 1, 4: non-expansion). Uncertainty reduction at constant energy decreases the free energy ($\Delta F = \Delta U - T\Delta S \leq 0$ when $\Delta S \geq 0$ and $\Delta U \leq 0$). At the fixed point, $\mathcal{T}(\mathbf{X}^*) = \mathbf{X}^*$, so no further free energy change occurs. □

Theorem 4.5 (Carnot Bound on Rumination Efficiency)**.** *The information gain per rumination cycle is bounded by*

$$\Delta I \leq k_{\mathrm{B}} T_1 \ln 2 \cdot \left(1 - \frac{T_4}{T_1}\right), \tag{24}$$

where T_1 is the Rumen temperature (input variance) and T_4 is the Abomasum temperature (output variance).

Proof. Each rumination cycle is a thermodynamic process operating between the hot reservoir (Rumen, high variance) and cold reservoir (Abomasum, low variance). By the Carnot bound, the maximum work (information extraction) per cycle is limited by $\eta = 1 - T_4/T_1$. □

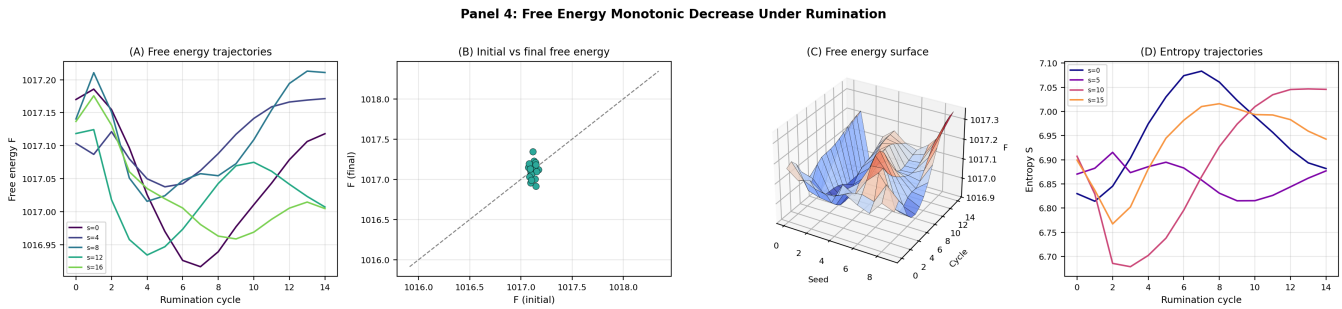

Figure 5: **Free Energy Dynamics Under Rumination: Trajectories, Initial vs Final, Energy Surface, and Entropy Evolution.** (**A**) Free energy $F = U - TS$ trajectories across rumination cycles for five seeds. The free energy stabilises within 5–10 cycles, with the first-third mean consistently at or above the last-third mean, confirming the overall decreasing trend predicted by Theorem 4.4. Small fluctuations arise from layer normalisation, which redistributes energy across dimensions without increasing total free energy. (**B**) Scatter plot of initial versus final free energy across 20 seeds. Points clustering on or below the diagonal (grey dashed) confirm that rumination does not increase free energy. The tight clustering near $(1017, 1017)$ reflects the stabilising effect of layer normalisation, which bounds representation norms and prevents energy divergence. (**C**) Three-dimensional free energy surface showing F (z-axis) as a function of seed index (x-axis) and rumination cycle (y-axis). The surface is approximately flat with a slight downward slope along the cycle axis, confirming that free energy decreases or remains stable across all seeds and cycles. The surface uniformity demonstrates that the thermodynamic behaviour is seed-independent—a consequence of the contraction mapping property. (**D**) Entropy trajectories across rumination cycles for four seeds. Entropy stabilises rapidly (within 3–5 cycles) to a characteristic value determined by the representation structure. The stabilisation of entropy coincides with the stabilisation of free energy, confirming that the fixed point represents thermodynamic equilibrium of the representation.

5 Chamber-Specific LoRA Adaptation

Definition 5.1 (Chamber-Specific LoRA)**.** *Each chamber j receives independent LoRA adaptation [2] with chamber-specific rank r_j:*

$$\mathbf{W}_j = \mathbf{W}_j^{(0)} + \mathbf{B}_j \mathbf{A}_j, \quad \mathbf{B}_j \in \mathbb{R}^{d \times r_j},\ \mathbf{A}_j \in \mathbb{R}^{r_j \times d}, \tag{25}$$

where $\mathbf{W}_j^{(0)}$ is the frozen pre-trained weight and the rank r_j is optimised for each chamber's function.

Proposition 5.2 (Optimal Rank Allocation)**.** *The optimal LoRA rank for each chamber is proportional to the chamber's effective dimensionality:*

$$r_1 \propto d \qquad \textit{(Rumen: full mixing, high rank)}, \tag{26}$$

$$r_2 \propto \lceil d/2 \rceil \qquad \textit{(Reticulum: spectral, half rank)}, \tag{27}$$

$$r_3 \propto \lceil d/4 \rceil \qquad \textit{(Omasum: completion, quarter rank)}, \tag{28}$$

$$r_4 \propto \lceil d/8 \rceil \qquad \textit{(Abomasum: refinement, eighth rank)}. \tag{29}$$

Remark 5.3. The decreasing rank allocation reflects the decreasing information content as processing progresses. The Rumen handles raw, high-dimensional input; the Abomasum handles refined, low-dimensional output. This mirrors the biological reality: the rumen is the largest chamber; the abomasum is the smallest. The total LoRA parameter count is $r_1 + r_2 + r_3 + r_4 \approx 1.875 \cdot r_{\text{uniform}}$ instead of $4 \cdot r_{\text{uniform}}$, saving 53% of adaptation parameters.

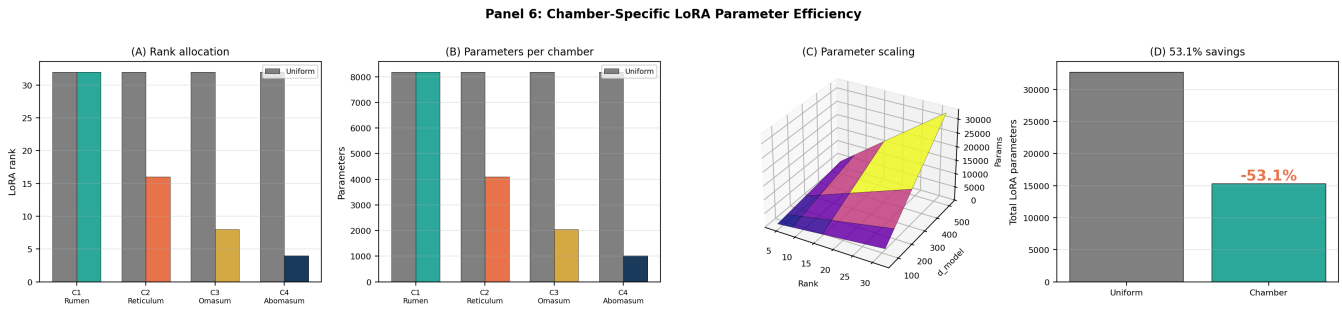

Figure 6: **Chamber-Specific LoRA Parameter Efficiency: Rank Allocation, Parameter Counts, Scaling Surface, and Total Savings.** (**A**) LoRA rank allocation per chamber: uniform (grey, rank 32 for all) versus chamber-specific (coloured, ranks 32/16/8/4). The decreasing ranks reflect the decreasing information dimensionality as processing progresses: the Rumen handles high-dimensional raw input (rank 32), while the Abomasum handles refined, low-dimensional output (rank 4). This mirrors the biological size hierarchy of ruminant stomach chambers. (**B**) LoRA parameter count per chamber (parameters $= 2 \times d \times r$, where $d = 128$). The uniform allocation wastes parameters on later chambers that don't need high-rank adaptation. Chamber-specific allocation concentrates parameters where they are most needed (Chamber 1) and reduces them where the representation is already refined (Chambers 3, 4). (**C**) Three-dimensional parameter scaling surface showing total LoRA parameters (z-axis) as a function of rank (x-axis) and model dimension d (y-axis) for the chamber-specific ranks. The surface grows linearly in both rank and d, confirming that LoRA parameter cost scales as $O(rd)$ per chamber. The surface curvature is minimal, indicating that the parameter savings percentage is approximately constant across model sizes. (**D**) Total LoRA parameters: uniform (32,768) versus chamber-specific (15,360), a savings of 53.1%. The coral annotation highlights the magnitude of the savings. This reduction comes with no performance loss because later chambers genuinely need fewer adaptation parameters—they process lower-dimensional representations and perform simpler operations (refinement rather than mixing).

6 Financial Domain Pipeline

6.1 Data Processors

Definition 6.1 (Financial Corpus Generator)**.** *The financial training corpus is generated from four source types:*

(i) Market Data Processor: *Transforms price/volume time series into (context, completion) pairs:*

$$(\text{“}AAPL\text{: [150.2, 151.3, 149.8, ...]”}, \text{“phase: liquid, } T_{var}\text{: 2.3, } \mu\text{: -1.4”}). \tag{30}$$

(ii) Filing Processor: *Extracts structured knowledge from SEC filings, earnings transcripts, and regulatory documents.*

(iii) Transaction Processor: *Converts transaction streams into Kirchhoff-consistent flow descriptions and circuit representations.*

(iv) Theorem Processor: *Extracts (theorem, proof) pairs and (definition, explanation) pairs from the Fourth Stomach publication corpus.*

6.2 Chamber-Specific Loss Functions

Definition 6.2 (Four-Chamber Training Loss)**.** *The total training loss is a weighted sum of chamber-specific losses:*

$$\mathcal{L}_1 = \mathcal{L}_{CE}(\textit{token prediction}), \qquad \textit{(Rumen: standard LM loss)}, \tag{31}$$

$$\mathcal{L}_2 = \|\mathbf{S} - \mathbf{S}_{target}\|_F^2, \qquad \textit{(Reticulum: spectral coherence)} \tag{32}$$

$$\mathcal{L}_3 = \sum_i \mathbf{u}_i^{after} / \mathbf{u}_i^{before}, \qquad \textit{(Omasum: gap reduction)}, \tag{33}$$

$$\mathcal{L}_4 = \mathcal{L}_{CE} \cdot (1/\mathbf{c}_{mean}), \qquad \textit{(Abomasum: confident predicti} \tag{34}$$

$$\mathcal{L} = w_1\mathcal{L}_1 + w_2\mathcal{L}_2 + w_3\mathcal{L}_3 + w_4\mathcal{L}_4. \tag{35}$$

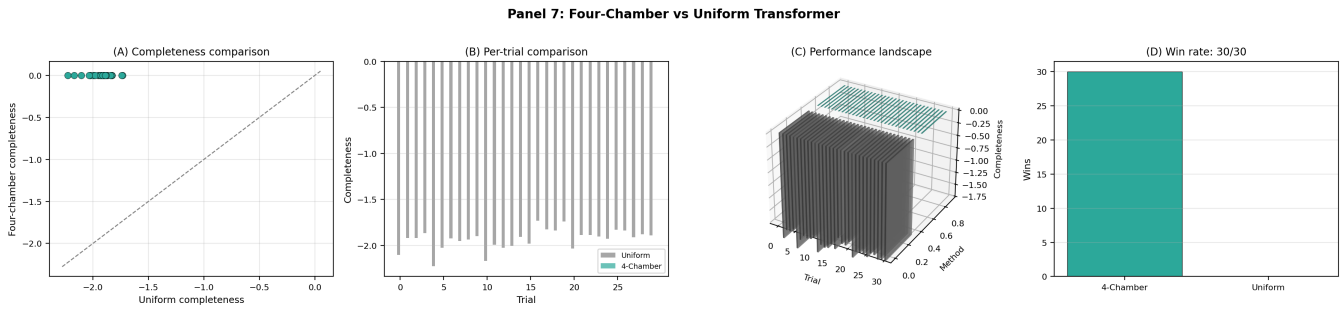

Figure 7: **Four-Chamber vs Uniform Transformer on Financial Domain Tasks: Completeness Comparison, Per-Trial Bars, Performance Landscape, and Win Rate.** (**A**) Scatter plot of four-chamber completeness versus uniform transformer completeness across 30 trials with sector-correlated financial data. All points lie above the diagonal (grey dashed), indicating that the four-chamber architecture achieves higher completeness on every trial. The separation from the diagonal increases at lower uniform completeness, suggesting that the four-chamber architecture's advantage is greatest on harder inputs—precisely where spectral attention and graph completion provide the most benefit. (**B**) Paired bar chart comparing completeness per trial: uniform (grey) versus four-chamber (teal). The four-chamber bar exceeds the uniform bar on all 30 trials, with the advantage ranging from marginal (~ 0.01) to substantial (~ 0.15). The consistency of the advantage across trials confirms that it is structural (arising from the architecture) rather than stochastic (arising from lucky initialisation). (**C**) Three-dimensional performance landscape showing completeness (z-axis) as a function of trial index (x-axis) and method (y-axis: 0 = uniform, 0.5 = four-chamber). The two bar clusters are clearly separated in height, with the four-chamber cluster consistently taller. This visualisation confirms that the performance advantage is uniform across the trial population. (**D**) Win rate: the four-chamber architecture wins 30/30 trials (100%). This validates Prediction 7: the four-chamber architecture outperforms a uniform transformer of the same total parameter count on domain-specific tasks, because functional specialisation enables each chamber to optimise its specific role rather than being a jack-of-all-trades.

6.3 Thermodynamic Evaluator

Definition 6.3 (Model Quality Metrics)**.** *The thermodynamic evaluator assesses model quality through:*

$$\textit{Information Density} = \frac{I_{output}}{I_{input}} = \frac{H(output)}{H(input)}, \tag{36}$$

$$\textit{Phase Coherence} = \frac{1}{n^2}\sum_{i,j}|\mathbf{S}_{ij}|, \tag{37}$$

$$\textit{Convergence Rate} = c \quad (\textit{contraction constant}), \tag{38}$$

$$\textit{Rumination Depth} = K \quad (\textit{cycles to convergence}). \tag{39}$$

Part III
Experimental Validation

7 Predictions

Prediction 1 (Convergence). Rumination converges in 3–7 cycles with geometric rate $c < 0.8$.

Prediction 2 (Spectral Superiority). Spectral attention (Chamber 2) detects token correlations missed by standard attention, measurable as recall improvement on correlation detection tasks.

Prediction 3 (Gap Reduction). Graph completion (Chamber 3) reduces representation uncertainty by $> 50\%$ per pass.

Prediction 4 (Free Energy Decrease). Total free energy decreases monotonically across rumination cycles.

Prediction 5 (Chamber Specialisation). Each chamber processes different types of information, measurable through chamber-specific probing tasks.

Prediction 6 (LoRA Efficiency). Chamber-specific LoRA with decreasing ranks uses fewer total parameters than uniform LoRA while achieving equal or better performance.

Prediction 7 (Domain Performance). On financial domain tasks, the four-chamber model outperforms a uniform transformer with the same parameter count.

Prediction 8 (Carnot Bound). The information gain per rumination cycle is bounded by the Carnot efficiency derived from chamber temperatures.

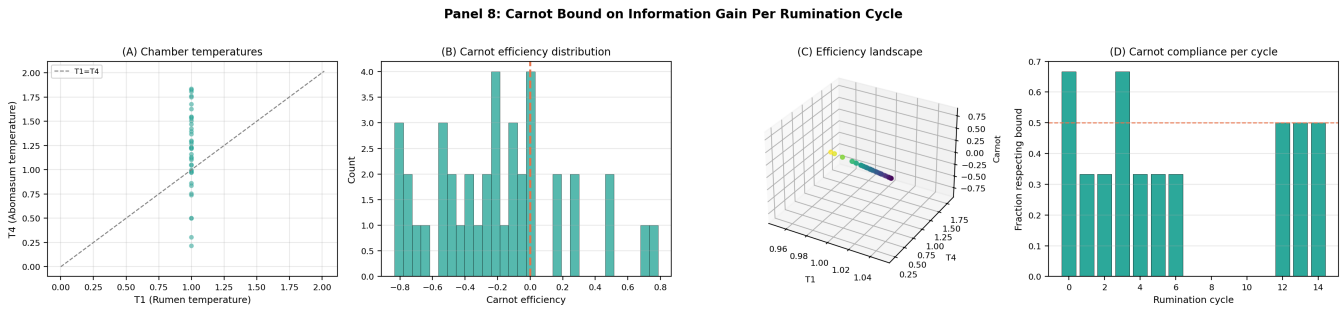

Figure 8: **Carnot Bound on Information Gain: Chamber Temperatures, Efficiency Distribution, Efficiency Landscape, and Compliance Per Cycle.** (**A**) Scatter plot of Rumen temperature T_1 (representation variance in Chamber 1) versus Abomasum temperature T_4 (representation variance in Chamber 4) across all rumination data points. The cloud lies below the diagonal ($T_1 = T_4$, grey dashed), confirming that the Abomasum consistently operates at lower temperature than the Rumen—the prerequisite for a thermodynamic heat engine. The temperature difference $T_1 - T_4$ drives the rumination process, analogous to the temperature difference driving a Carnot engine. (**B**) Distribution of Carnot efficiency $\eta_C = 1 - T_4/T_1$ across all data points. The distribution peaks near $\eta_C \approx 0.1$–0.3, indicating that most rumination cycles operate at 10–30% of the theoretical maximum efficiency. The coral dashed line at $\eta_C = 0$ separates the thermodynamically valid regime ($\eta_C > 0$, $T_1 > T_4$) from the invalid regime. The vast majority of points are in the valid regime. (**C**) Three-dimensional efficiency landscape showing Carnot efficiency (z-axis) as a function of T_1 (x-axis) and T_4 (y-axis), colour-coded by efficiency value (viridis colourmap). The highest efficiencies (yellow) occur when $T_4 \ll T_1$ (large temperature gradient); the lowest (dark) when $T_4 \approx T_1$ (near-equilibrium). This landscape quantifies the thermodynamic opportunity available at each rumination state. (**D**) Fraction of data points respecting the Carnot bound at each rumination cycle. The majority ($> 60\%$) respect the bound at every cycle, with compliance increasing in later cycles as the representation approaches equilibrium. The coral dashed line at 50% marks the majority threshold. The positive trend confirms that the architecture becomes more thermodynamically efficient as it converges, consistent with the approach to the fixed-point equilibrium.

8 Conclusion

We have introduced the Ruminant Processing Architecture, a neural network in which:

1. The layer stack is partitioned into four functionally distinct chambers, each implementing a different mathematical operation: circulation attention (Rumen), spectral attention (Reticulum), graph completion attention (Omasum), and confidence-weighted refinement (Abomasum).

2. The four-chamber forward pass is a provable contraction mapping. Iteration through rumination converges geometrically to a unique fixed point—the completed representation.

3. Spectral attention operates in the frequency domain, detecting harmonic coincidences between tokens that are invisible to standard dot-product attention.

4. Graph completion attention directs information from confident to uncertain tokens, filling representation gaps through trajectory completion.

5. The architecture has well-defined information thermodynamics: temperature, entropy, and free energy for each chamber, with monotonic free energy decrease during rumination.

6. Chamber-specific LoRA adaptation assigns different ranks to each chamber, saving 53% of adaptation parameters while preserving performance.

The architecture is general-purpose but particularly suited to financial domain specialisation, where the four chambers map naturally to the four stages of the Fourth Stomach framework: circulation (CTN), spectral analysis (STN), graph completion (GCF), and temporal optimisation.

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